

CURRICULUM VITAE

Date Prepared: 08/25/2026

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Education:

09/2012– 06/2016	B.S.	Mechanical Engineering	Huazhong University of Science and Technology Wuhan, China
09/2016– 12/2022	Ph.D.	Mechatronic Engineering (Thesis advisor: Jianfeng Xu, Ph.D.)	Huazhong University of Science and Technology Wuhan, China

Postdoctoral Training:

09/2023– present	Research Fellow	Nuclear Medicine and Molecular Imaging; PET/SPECT detector instrumentation (Primary mentor: Hamid Sabet, Ph.D.)	Massachusetts General Hospital / Harvard Medical School Boston, MA
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Appointments at Hospitals/Affiliated Institutions:

09/2023– present	Research Fellow	Department of Radiology (Division of Nuclear Medicine and Molecular Imaging)	Massachusetts General Hospital Boston, MA
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Other Professional Positions:

09/2017– 03/2018	Visiting Scholar (Supervisors: Qiyu Peng, Ph.D.; Thomas F. Budinger, M.D., Ph.D.)	Molecular Biophysics and Integrated Bioimaging, Lawrence Berkeley National
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09/2020– 09/2021	Visiting Scholar (Supervisors: Qiyu Peng, Ph.D.)	Laboratory, Berkeley, CA Institute of Biomedical Engineering, Shenzhen Bay Laboratory, Shenzhen, China
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Editorial Activities:

- **Other Editorial Roles**

2024– present	Guest Editor	Symmetry (MDPI, Basel, Switzerland) Special Issue: Symmetric/Asymmetric Study in Medical Imaging
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Honors and Prizes:

2013	Merit Student	Huazhong University of Science and Technology	Academic excellence
2014	First Prize, "Challenge Cup" National College Student Business Plan Competition	National "Challenge Cup" Organizing Committee, China	National student innovation and entrepreneurship competition
2014	National Encouragement Scholarship	Ministry of Education, China	Academic achievement
2016	Outstanding Graduate	Huazhong University of Science and Technology	Undergraduate academic distinction
2017	National Scholarship (Graduate)	Ministry of Education, China	Graduate academic and research achievement
2021	Merit Postgraduate Student	Huazhong University of Science and Technology	Graduate academic distinction
2022	Distinguished Graduate	Huazhong University of Science and Technology	Doctoral academic distinction

Report of Funded and Unfunded Projects

Past

08/2007– 03/2019	Exploring the Benefits of Time-of-Flight PET National Institutes of Health, R01EB006085 Visiting graduate researcher (PI: Qiyu Peng) Developed and characterized time-of-flight PET detector modules with sub-150 ps coincidence resolving time for the Tachyon-II scanner.
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12/2018–
11/2025 High-Performance SPECT System for Cardiac Imaging

National Institutes of Health, R01HL145160
Research Fellow (postdoctoral researcher) (PI: Hamid Sabet)
Designed detector-module electronics and FPGA-based readout for a high-sensitivity, high-resolution SPECT system for cardiac imaging.

Current

09/2023–
08/2027 High-Performance SPECT for Dynamic Brain Imaging

National Institutes of Health, R01EB034785
Research Fellow (postdoctoral researcher) (PI: Hamid Sabet)
Develops laser-processed scintillator detectors and FPGA-based readout electronics enabling dynamic brain SPECT imaging with improved sensitivity and spatial resolution.

Projects Submitted for Funding

Pending;
Council review
October 2026 High-Resolution Laser-Engineered Continuous-Crystal PET: From Small Animal to Brain Imaging

National Institutes of Health / NIBIB, K99/R00 (1K99EB040542-01)
Principal Investigator
Develops laser-engineered continuous-crystal PET detectors with high spatial resolution and depth-of-interaction capability and translates the technology from small-animal imaging toward human brain imaging.
Reviewed by ZRG1 T15-E on 07/16/2026; impact score 27.

Report of Scholarship

ORCID iD: 0000-0002-4918-0810

Peer-Reviewed Scholarship in print or other media:

Research Investigations

1. Xie S*, **Zhang X***, Peng H, et al. PET detectors with 127 ps CTR for the Tachyon-II time-of-flight PET scanner. Nucl Instrum Methods Phys Res A. 2019;933:48-55. doi:10.1016/j.nima.2019.03.083 (*co-first authors)
2. **Zhang X**, Xie S, Yang J, et al. A depth encoding PET detector using four-crystals-to-one-SiPM coupling and light-sharing window method. Med Phys. 2019;46(8):3385-3398. doi:10.1002/mp.13603
3. Xu J, Xie S, **Zhang X**, et al. A preclinical PET detector constructed with a monolithic scintillator ring. Phys Med Biol. 2019;64(15):155009. doi:10.1088/1361-6560/ab2ca4
4. Xie S, Xu J, Yang M, Ying G, **Zhang X**, et al. Methods to improve light transport efficiency in LYSO crystals based on characteristics of optical reflectance. IEEE Trans Nucl Sci. 2019;66(9):2100-2106. doi:10.1109/TNS.2019.2931929
5. Zhao Z, Xie S, **Zhang X**, et al. An advanced 100-channel readout system for nuclear imaging. IEEE Trans Instrum Meas. 2019;68(9):3200-3210. doi:10.1109/TIM.2018.2877952
6. Song Z, Zhao Z, Yu H, Yang J, **Zhang X**, et al. An 8.8 ps RMS resolution time-to-digital converter implemented in a 60 nm FPGA with real-time temperature correction. Sensors. 2020;20(8):2172. doi:10.3390/s20082172

7. Xie S, **Zhang X**, Huang Q, Gong Z, Xu J, Peng Q. Methods to compensate the time walk errors in timing measurements for PET detectors. *IEEE Trans Radiat Plasma Med Sci.* 2020;4(5):555-562. doi:10.1109/TRPMS.2020.2981388
8. Xie S, **Zhang X**, Zhang Y, et al. Evaluation of various scintillator materials in radiation detector design for positron emission tomography (PET). *Crystals.* 2020;10(10):869. doi:10.3390/cryst10100869
9. **Zhang X**, Ye B, Yu H, et al. Depth of interaction measurements based on rectangular light sharing window technology and nine-crystals-to-one-SiPM coupling method. *IEEE Trans Radiat Plasma Med Sci.* 2021;5(3):306-314.
10. **Zhang X**, Xie Q, Xie S, Yu X, Xu J, Peng Q. A novel portable gamma radiation sensor based on a monolithic lutetium-yttrium oxyorthosilicate ring. *Sensors.* 2021;21(10):3376. doi:10.3390/s21103376
11. **Zhang X**, Yu H, Xie Q, et al. Design study of a PET detector with 0.5 mm crystal pitch for high-resolution preclinical imaging. *Phys Med Biol.* 2021;66(13):135013. doi:10.1088/1361-6560/ac0b82
12. Xie S, Zhu Z, **Zhang X**, et al. Optical simulation and experimental assessment with time-walk correction of TOF-PET detectors with multi-ended readouts. *Sensors.* 2021;21(14):4681. doi:10.3390/s21144681
13. **Zhang X**, Yu X, Zhu Z, et al. Development and evaluation of a dual-layer-offset PET detector constructed with different reflectors. *Crystals.* 2022;12(1):93. doi:10.3390/cryst12010093
14. Yu X, **Zhang X**, Zhang H, et al. Requirements of scintillation crystals with the development of PET scanners. *Crystals.* 2022;12(9):1302. doi:10.3390/cryst12091302
15. **Zhang X**, Yu X, Zhang H, et al. Development and evaluation of 0.35-mm-pitch PET detectors with different reflector arrangements. *IEEE Trans Radiat Plasma Med Sci.* 2023;7(8):802-809. doi:10.1109/TRPMS.2023.3307128
16. Cheng H, Liu R, Fang S, Li Z, Zhang D, **Zhang X**, et al. Synthesis of easily-modified and useful dibenzo[b,d]azepines by palladium(II)-catalyzed cyclization/addition with a green solvent. *Chem Commun.* Published online March 26, 2024. doi:10.1039/D3CC06321F
17. Cheng H, Bai L, **Zhang X**, et al. 68Ga labeled olmutinib: design, synthesis, and evaluation of a novel PET EGFR probe. *Bioorg Chem.* 2024;153:107987. doi:10.1016/j.bioorg.2024.107987
18. Yu X, **Zhang X**, Zeng J, et al. Depth of interaction in PET detector design: performance optimization with light-sharing window. *IEEE Trans Instrum Meas.* 2025;74:1-10. doi:10.1109/TIM.2024.3522335
19. **Zhang X**, Yu X, Cheng H, et al. Enhancing gamma-ray detection: processing grooved microstructures on LYSO crystal with femtosecond laser. *Med Phys.* 2025;52(4):2644-2660. doi:10.1002/mp.17665

Thesis:

Zhang X. High-precision Positioning of Gamma Photons in Positron Emission Tomography Detectors. Wuhan, China: Huazhong University of Science and Technology; 2022.

Abstracts, Poster Presentations, and Exhibits Presented at Professional Meetings:

1. **Zhang X**, Sitek A, Blackberg L, Kupinski M, Furenlid L, Sabet H. Development and performance evaluation of a laser processed CsI:Tl detector with converging pixels. In: 2025 IEEE Nuclear Science Symposium (NSS), Medical Imaging Conference (MIC) and Room Temperature Semiconductor Detector Conference (RTSD). 2025:1-1. doi:10.1109/NSS/MIC/RTSD57106.2025.11286546
2. **Zhang X**, Feng Y, Ottensmeyer MP, Sabet H. Modular and robust PET platform for evaluating detector configurations. In: 2025 IEEE Nuclear Science Symposium (NSS), Medical Imaging Conference (MIC) and Room Temperature Semiconductor Detector Conference (RTSD). 2025:1-1. doi:10.1109/NSS/MIC/RTSD57106.2025.11287227

3. Hashemi A, **Zhang X**, Kouame D, Sabet H. Enhancing detector event positioning data using L1 regularization. In: 2025 IEEE Nuclear Science Symposium (NSS), Medical Imaging Conference (MIC) and Room Temperature Semiconductor Detector Conference (RTSD). 2025:1-2.
doi:10.1109/NSS/MIC/RTSD57106.2025.11287406

Narrative Report

I design the detectors that determine what a nuclear medicine scanner can see. In positron emission tomography (PET) and single photon emission computed tomography (SPECT), a patient receives a trace amount of a radioactive compound, and the scanner reconstructs where it travelled by catching the gamma rays it emits. Almost everything that matters clinically — whether a small lesion is visible, whether a drug reaching the brain can be measured — is set by the detector: the crystal that converts a gamma ray into light, the sensor that collects that light, and the electronics that measure it. My contribution has been to treat that chain as one system rather than as three separate components, and to remove the manufacturing limits that have constrained it for two decades.

My early work addressed the geometry of the crystal. Conventional detectors are built by cutting a crystal into thousands of tiny pixels and gluing them back together, which caps how fine the image can be and how much of the signal survives. I developed light-sharing and depth-encoding methods that let one sensor read several crystal elements at once and still recover where, and at what depth, each gamma ray interacted — the depth information being what keeps images sharp at the edge of the field of view, where lesions are most often missed. Applying these methods, I built a miniature preclinical scanner from continuous crystal rings that achieved 0.94 mm resolution, and designed detectors with crystal elements as small as a quarter of a millimetre. In parallel I rebuilt the electronics: a 100-channel readout system that made large detector arrays affordable. This work appears in *Physics in Medicine and Biology*, *Medical Physics*, and the *IEEE Transactions on Radiation and Plasma Medical Sciences*.

A visiting appointment at Lawrence Berkeley National Laboratory shaped how I think about this work. Learning from Dr. Thomas Budinger, then in his eighties and still at the bench, taught me that contributions of consequence in this field are measured over careers rather than projects, and that physics only matters here once it reaches a patient.

Since joining Massachusetts General Hospital I have pursued what I believe is now the binding constraint in high-performance PET: not the electronics or the algorithms, which my field has largely solved, but the fact that detector shapes are limited to what can be sawn and assembled by hand. My approach is to write the structure inside the crystal instead of cutting it — using a femtosecond laser to form optical barriers within an intact crystal, so that a single solid block behaves as though it were finely segmented. This makes detector geometries possible that cannot be manufactured any other way, including the large converging-pixel detector I built and characterized at MGH; I also showed that laser-formed surface structures can replace slower and costlier chemical and mechanical processing, which matters for whether the technology can be produced at scale. Alongside this, I lead the detector electronics for two NIH-funded SPECT systems for cardiac and dynamic brain imaging, and I built a modular PET test platform that lets a new detector concept be evaluated by swapping one module rather than building an entire scanner — work intended from the outset to be used by others.

My aim over the next several years is to carry laser-engineered continuous-crystal detectors from the bench into a brain PET system with millimetre-scale resolution, so that the earliest changes of neurodegenerative disease can be seen and measured rather than inferred. I collaborate on this with colleagues at Harvard, Yale, the University of California, Davis, and Cornell, and serve the field as a guest editor and as a reviewer for journals in medical imaging and instrumentation.